



Relax and Recover

BAREOS with Relax and Recover (rear) Workshop



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Introduction to Relax and Recover (rear)

- Proven solution at large enterprise customers
- Rear established as standard solution for Linux disaster recovery in data centers
- Shipping with Fedora and openSUSE
- Integrates with many “commercial” backup software solutions, e.g. TSM, DP, NBU, NSR, ...
- Integrates with many “open source” backup software solutions, e.g. BAREOS, BACULA, RBME
- Integrates with OS backup software solutions as well, e.g. GNU tar, rsync, ...
- Scales well with large amounts of servers



Rear Features

- Focus on disaster recovery and not backup
- Tight integration with common backup software
- Simple full backup integrated
- Complements backup software
 - Backup software: data storage and retrieval
 - Rear: recover the system layout and make it work
 - Rear: use the backup software to restore data
- Methodology: use the best tool for the job



Usage of rear

- Shell scripts are stored under `/usr/share/rear`
- Scripts are kept together according work-flows
 - mkrescue (only make rescue image)
 - mkbackup (including make rescue image)
 - mkbackuponly (excluding make rescue image)
 - recover (the actual recovery part)
- Easy to incorporate new scripts, e.g. for information gathering of Hard- and Software, or other goodies



Decide on DR strategy

- Which backup mechanism to use?
 - GNU tar, rsync, bacula, bareos, commercial backup program
- Where will the backups reside?
 - NFS share, CIFS share, external USB disk, tape, local (spare) disk
 - Remote network location
- How shall we start the rescue image
 - Via CDROM (ISO image), tape (OBDR), network (PXE), USB disk



Backup Types

- The major “[backup types](#)” available are
 - **NETFS:** NFS, CIFS, USB, TAPE, ISO, SSHFS
 - **RSYNC:** rsync method
 - **REQUESTRESTORE, EXTERNAL**
 - **BACULA, BAREOS, RBME** (open source backup software)
 - **DP, NBU, TSM, NSR, GALAXY[7], SESAM** (commercial stuff)
 - **DUPLICITY** (duplicity and/or duply)



Use Bareos as backup/restore with rear

- If your Linux client is fully back-up'ed with BAREOS
 - No need to take full TAR archive
 - Just create rescue image with bareos tools incl.
 - **rear mkrescue**
- Put the following in /etc/rear/site.conf (or local.conf)
 - OUTPUT=ISO
 - BACKUP=BAREOS
- Recover your system via booting up rescue image
 - **rear recover**



Disaster Recovery in Practice

- Gather system information
- Store the disk layout
 - Partitioning, LVM and RAID configuration
 - File systems, file system labels ...
 - Boot loader (GRUB, LILO, ELILO)
- Make a system backup (OS and user data)
- Create boot-able rescue media with system configuration (and optional with backup data)
- All steps are done “online”



Disaster Recovery: rescue media

- Create “rescue linux” from running system
- Optimally compatible “tool box”
- Clone the system environment
 - Linux kernel and modules
 - Device driver configuration
 - Network configuration
 - Basic system software and tools
- Operate entirely in RAM (initrd)



Rear mkrescue

- Will create an ISO image stored as
 - `/var/lib/rear/output/rear-$(hostname).iso`
 - On NFS server as `/path/$(hostname)/rear-\ $(hostname).iso`
- Inspect file `/var/lib/rear/layout/disklayout.conf`
- Try to boot from the ISO image into the RESCUE system
 - Use 'dmesg' to check if devices were found



Recovery Process in detail

- Boot system from rescue media
- Restore disk layout
 - Create partitions, RAID configuration and LVM
 - Create file systems (mkfs, mkswap)
 - Configure file systems (labels, mount points)
- Restore the backup data
- Restore the boot loader
- Inspect & Reboot



Recover with rear

- Boot rescue image and select 'recover'

```
Relax and Recover v1.13.0

Recover fedora

Other actions
Help for Relax and Recover
Boot First Local disk (hd0)
Boot Second Local disk (hd1)
Boot Next device
Hardware Detection Tool
ReBoot system
Power off system

Press [Tab] to edit, [F2] for help, [F1] for version info

Rear rescue image kernel 3.1.7-1.fc16.i686.PAE Thu, 03 May 2012 14:46:
BACKUP=NETFS OUTPUT=ISO BACKUP_URL=nfs://192.168.1.100/test
```



Contacts

Web-site: <http://relax-and-recover.org/>

GitHub: <https://github.com/rear/rear>

Mailing list: rear-users@lists.relax-and-recover.org

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